```python
import numpy as np
import matplotlib.pylab as plt



# b


def X(U, alpha):
return np.sqrt(-alpha*np.log(1-U))


def test_x(n, alpha):
U = np.random.uniform(size=n)
x = X(U, alpha)
plt.hist(x, bins=100, density=True)
plt.show()

# test_x(10_000_000, 1)

def Y():
U = np.random.uniform(size=5)
x = np.array([X(U[0], 1), X(U[1], 1), X(U[2], 1), X(U[3], 1.2), X(U[4], 1.2)])
x.sort()
return x[2]




def test_c(n):
Y_arr = np.empty(n)
for i in range(n):
Y_arr[i] = Y()
plt.hist(Y_arr, bins=100, density=True)
plt.show()


def P_Y_big_or_eq_v(v, n):
s = 0
for _ in range(n):
if Y() >= v:
s += 1
return s/n


print(P_Y_big_or_eq_v(1, 100000))


# P(Y >= 1 | Y >= 0.75)
def oppg_c_2(n):
return P_Y_big_or_eq_v(1, n)/P_Y_big_or_eq_v(0.75, n)
```

```
def oppg_c_3(n):
Y_vals = np.zeros(n)
for i in range(n):
Y_vals[i] = Y()
```